FAA FAARFIELD CM Report — Computational Mechanics

Print Report

FAARFIELD 2.1.1 (Build 03/18/2026)

Job: New Job 1 | **Structure:** New Structure 1 | **Analysis:** HMA on Aggregate | **Generated:** 2026-03-18 16:04:14

MAX TOTAL CDF	DESIGN THICKNESS	AIRCRAFT IN MIX	CRITICAL OFFSET	SUBGRADE MODULUS	CONVERGED / ITERATIONS
1.000000	**20.00** in.	**1**	**30** in.	**7,500** psi	**Yes / 1**

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A Pavement Structure Summary

Design Layers

Layer #	Thickness (in.)	Modulus (psi)	LCode
1	4.00	200,000	1
2	6.00	49,126	6
3	20.00	17,024	8
4	Semi-infinite	7,500	4

Expanded Sublayer Structure (after modulus adjustment)

Layer #	Thickness (in.)	Modulus (psi)	LCode
1	4.00	200,000	1
2	6.00	49,126	6
3	4.00	19,603	8
4	8.00	19,603	8
5	8.00	13,155	8
6	Semi-infinite	7,500	4

Evaluation Depth at Subgrade = 30.00 in.

Unbound Aggregate Modulus — Sublayering Procedure

Unbound aggregate layers (crushed base and uncrushed subbase) do not have a single fixed modulus. FAARFIELD subdivides each aggregate layer into sublayers and computes a depth-dependent modulus for each using an empirical formula. The modulus of each sublayer depends on the modulus of the material below it — sublayers near the bottom (close to the subgrade) have lower moduli, while sublayers near the top have higher moduli. The computation proceeds from the bottom of the aggregate layer upward.

Sublayer Modulus Reduction Formula

$$f_1 = 1 + C \times \ln(t) \,/\, \ln(10)$$

$$f_2 = D \times \ln(E_{i-1}) \times \ln(t) \,/\, \ln^2(10)$$

$$E_i = E_{i-1} \times (f_1 - f_2)$$

where t = sublayer thickness (in.), E_{i-1} = modulus of the layer below (psi). Applied iteratively from the bottom sublayer upward.

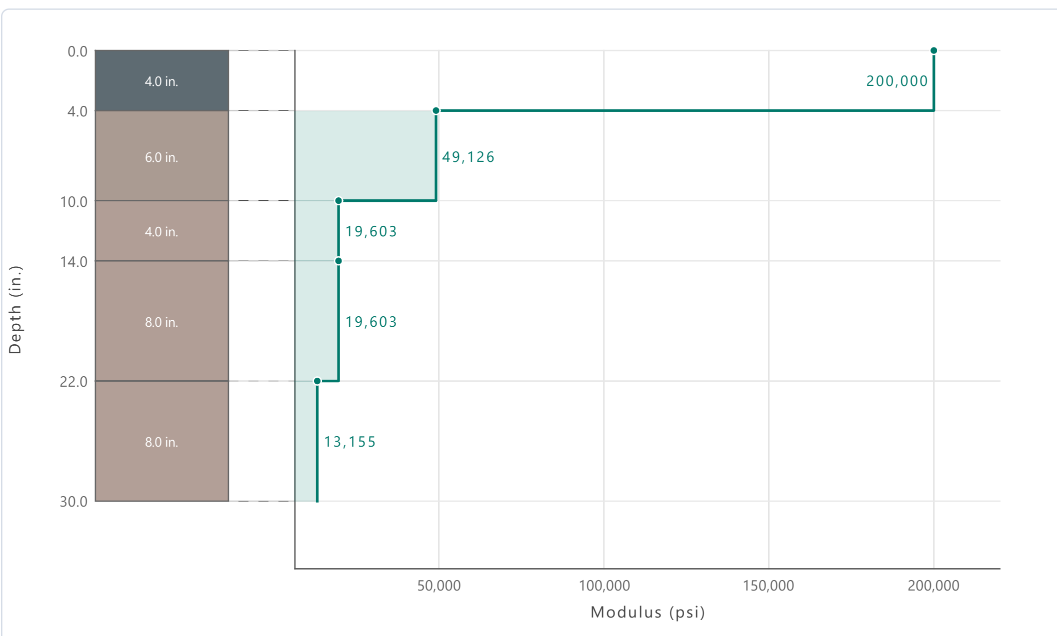
Parameter	P-209 Crushed Base	P-154 Uncrushed Subbase
C (thickness factor)	10.52	6.88
D (modulus interaction)	2.00	1.56
E_{below} (psi)	17,024	7,500
Number of sublayers	1	3

P-209 Crushed Aggregate Base — Sublayer Moduli

Sublayer	Thickness (in.)	Modulus (psi)
1 (top)	6.00	49,126
↓ *Layer below*	—	*17,024*

P-154 Uncrushed Aggregate Subbase — Sublayer Moduli

Sublayer	Thickness (in.)	Modulus (psi)
1 (top)	4.00	19,603
2	8.00	19,603
3 (bottom)	8.00	13,155
↓ *Layer below*	—	*7,500*



Modulus vs. depth profile for the expanded sublayer structure. Aggregate layers are subdivided and their moduli computed bottom→up using the empirical reduction formula. The teal step line traces the modulus at each sublayer depth.

B Design Equations

Subgrade Strain Failure Model (FAA Standard)

$$AA = 0.000247 + 0.000245 \times \log_{10}(E_{subgrade})$$

$$BB = 0.0658 \times E_{subgrade}^{0.559}$$

$$N_{fail} = 10{,}000 \times (AA / \varepsilon_v)^{BB}$$

For this structure ($E_{subgrade}$ = 7,500 psi): AA = 0.001196, BB = 9.647

Cumulative Damage Factor (CDF)

$$CDF_{aircraft} = \text{Repetitions} \times (C/P) / N_{fail}$$

$$CDF_{total} = \Sigma\, CDF_{aircraft} \quad \text{(summed over all aircraft)}$$

where C/P = Coverage-to-Pass ratio from Gaussian lateral wander model (σ = 30.435 in.)

Coverage-to-Pass (Gaussian Wander Model)

$$C/P(\text{offset}) = \int G(x; \sigma)\, dx \quad \text{over tire contact width}$$

$$G(x; \sigma) = [1 / (\sigma\sqrt{(2\pi)})] \times \exp(-x^2 / 2\sigma^2)$$

σ = 30.435 in. (std. dev. of lateral wander)

Projected tire width at depth d: TW_{proj} = TW + 2d

Convergence Criterion

$$|\ln(CDF_{total})| < 0.005 \quad \text{(exit criterion)}$$

Design converges when 0.995 < CDF_{total} < 1.005

Sublayers activated when 0.5 < CDF < 2.0 (|ln(CDF)| < 0.690)

C Understanding Coverage-to-Pass (C/P)

The Coverage-to-Pass (C/P) ratio is the probability that a given evaluation strip (10 in. wide) on the pavement surface will be loaded by a passing wheel, accounting for Gaussian lateral wander of the aircraft about the nominal wheel path centerline. The standard deviation of wander is σ = 30.435 in. (corresponding to a wander width of 70 in.).

For multi-wheel gear configurations, the total C/P at each strip is the superposition (sum) of the individual GaussArea contributions from every wheel in the gear assembly, because the entire gear wanders as a rigid body. FAARFIELD evaluates 41 strips on each side of the nominal wheel path centerline (offsets 0 to 400 in.) for a total of 81 bilateral offsets.

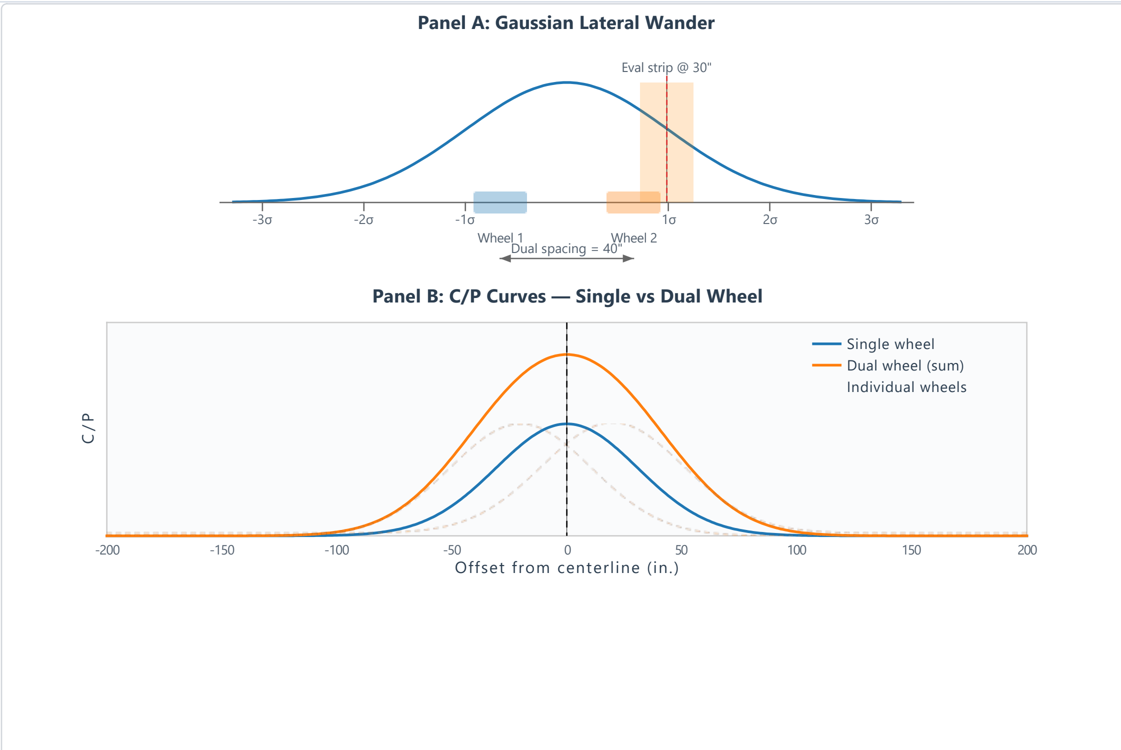


Figure 1: Gaussian lateral wander and single vs. dual wheel C/P comparison

D Fatigue Characterization

The following chart shows the subgrade fatigue model curve (allowable repetitions vs. vertical strain) with each aircraft's computed strain and N_{fail} plotted as scatter points.

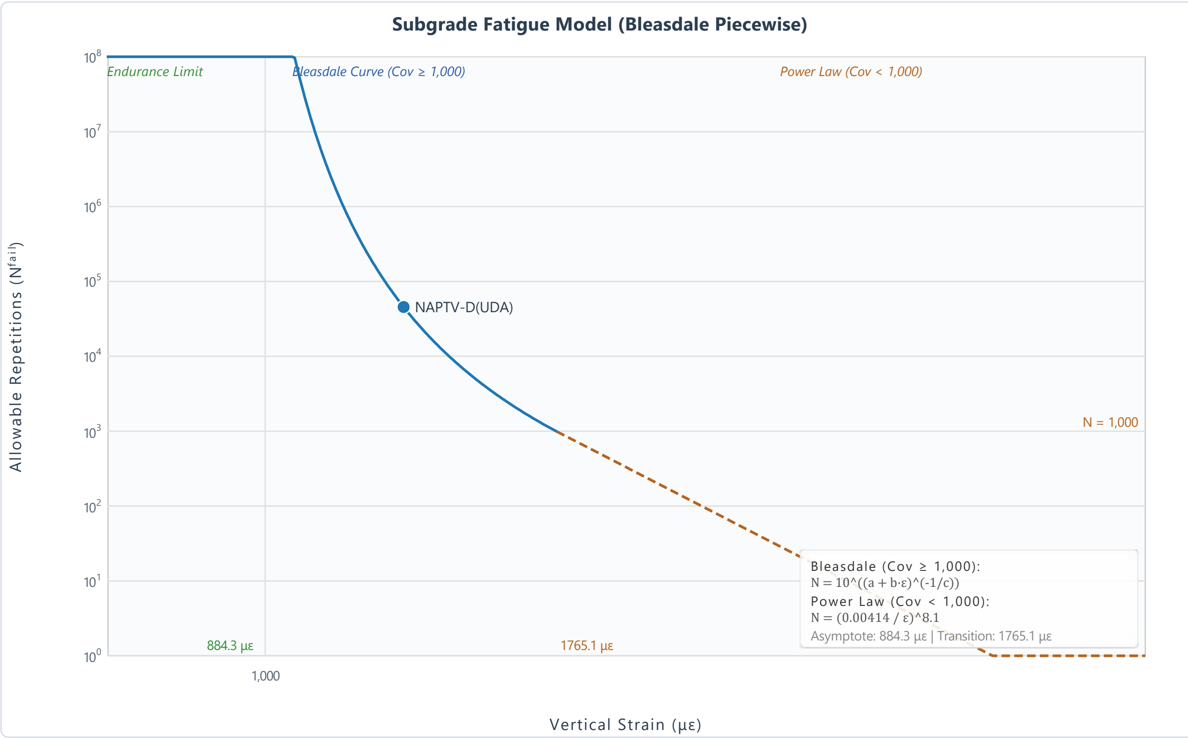


Figure 2: Subgrade fatigue model with aircraft scatter points

Aircraft Fatigue Parameters

Aircraft	Vert. Strain (με)	AA	BB	N_{fail}	Repetitions	N_{fail}/Reps	Model
NAPTV-D(UDA)	1310.46	0.004000	8.100	4.545E+04	89,569	5.07E-01	Bleasdale

Figure 3: Fatigue life reserve ratio per aircraft

E Per-Aircraft Detailed Breakdown

Aircraft 1: NAPTV-D(UDA)

Parameter	Value	Description
Gear Type	X	Landing gear configuration
Gear Load	72,000 lbs	Maximum gear load applied to pavement
Tire Pressure	200.0 psi	Inflation pressure
Tire Width (TW)	19.15 in.	Contact width at surface
Contact Area	360.00 in.²	Static tire contact area
Annual Departures	4,500	Annual departure level (ADL)

Parameter	Value	Description
Total Repetitions	89,569	Computed from ADL × 20-year design life
# Gear Loads	1	Main + belly gear load groups
Projected TW at Subgrade	79.16 in.	TW + 2×depth (45° spread)
Max Vertical Strain	1310.46 με	LEAF-computed subgrade strain
N_{fail}	4.545E+04	Allowable repetitions (Bleasdale)
Max C/P Ratio	0.50740	Peak coverage-to-pass ratio
Max CDF (this aircraft)	1.000000	Peak damage contribution
CDF at Critical Offset	1.000000	Damage at critical strip

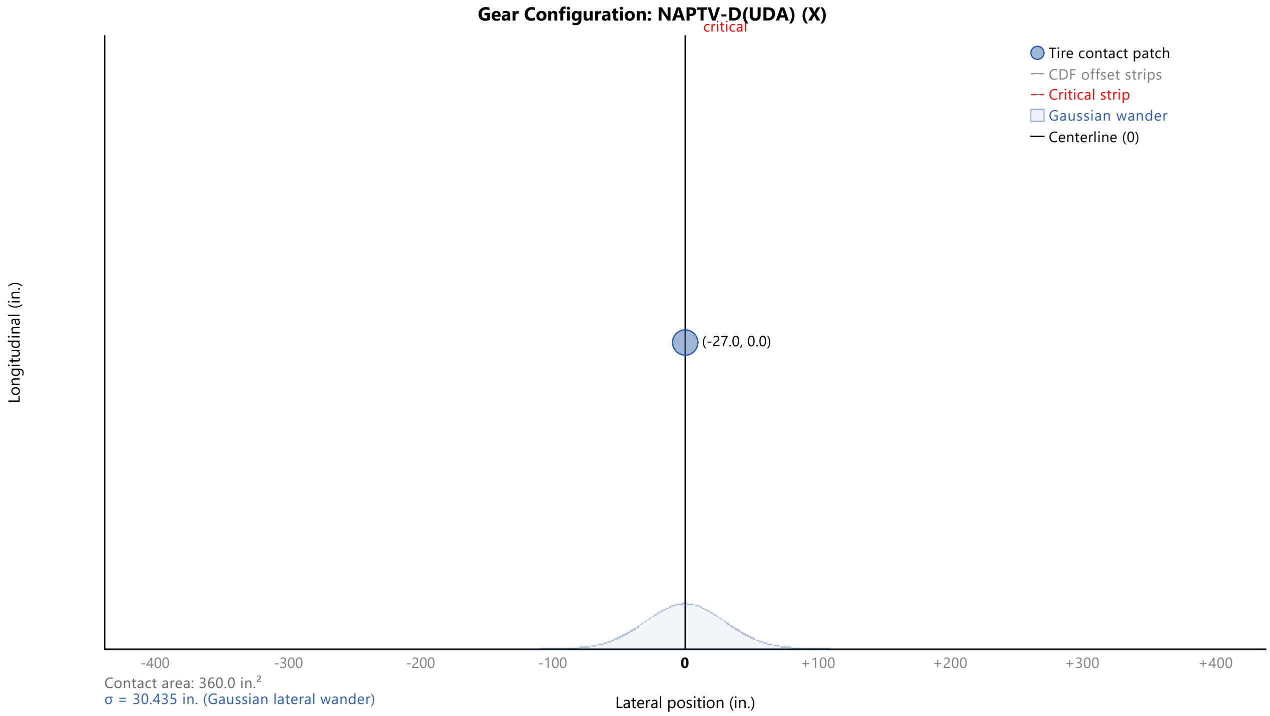


Figure 4: Gear configuration for NAPTV-D(UDA)

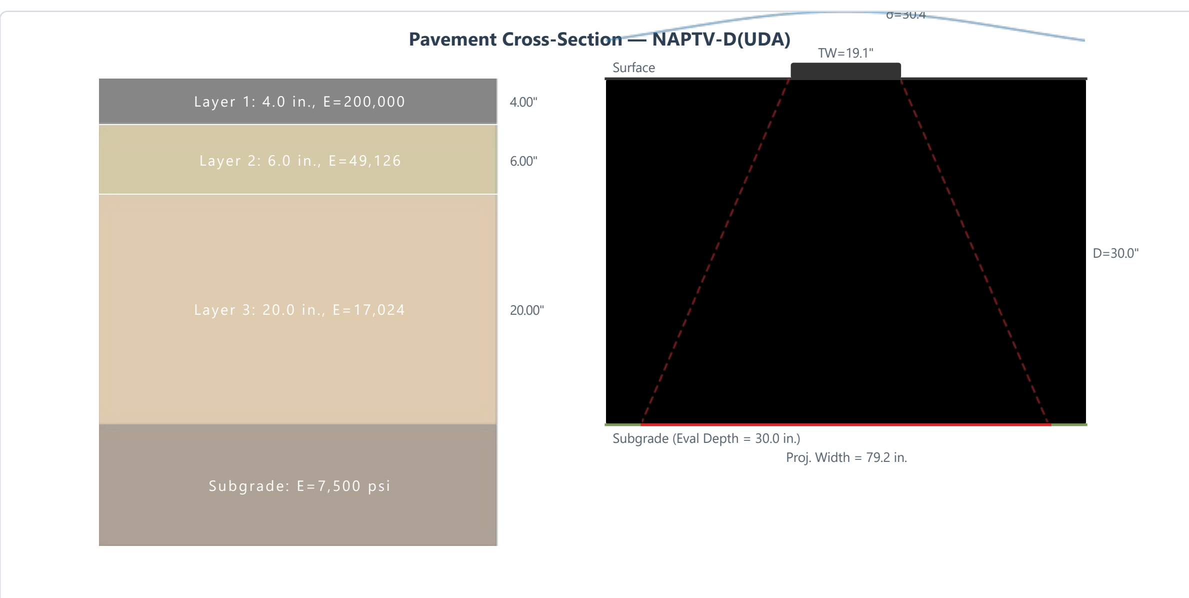


Figure 5: Pavement cross-section for NAPTV-D(UDA)

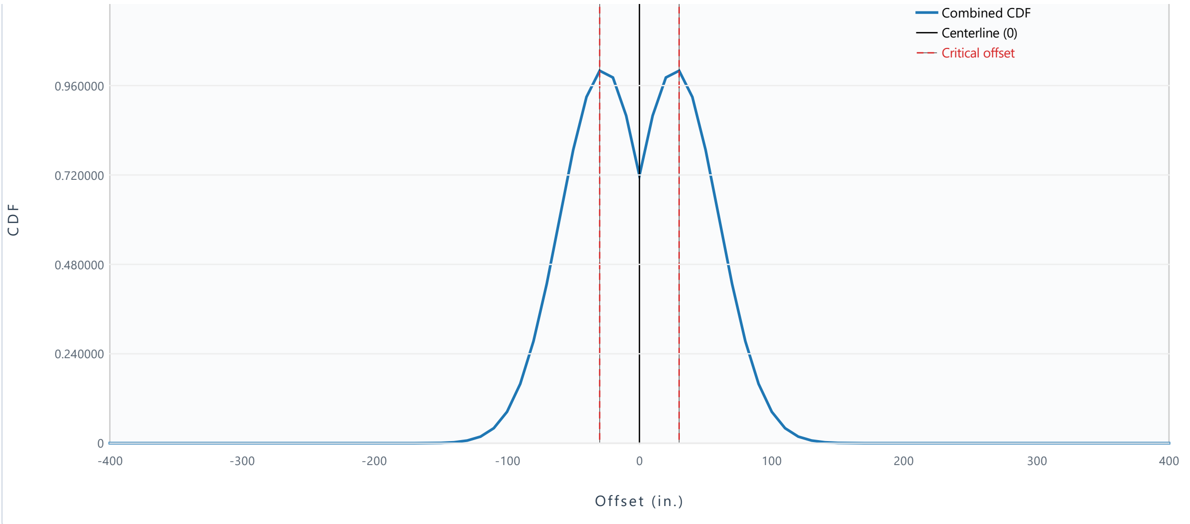


Figure 6: CDF by offset for NAPTV-D(UDA)

▶ CDF by Offset — NAPTV-D(UDA) (click to expand)

Step-by-Step Computation

1. Apply gear load of 72,000 lbs with tire pressure 200.0 psi and tire contact width 19.15 in. (X gear).
2. Run LEAF to compute vertical subgrade strain. Result: ε_v = 1310.46 με at evaluation depth 30.00 in..
3. Compute N_{fail} using Bleasdale model: AA=0.004000, BB=8.100. N_{fail} = 4.545E+04.